



< News > : A recent study claims that drinking lemon juice cures cancer without chemotherapy.

Step 1. Veracity Context-Aware Knowledge Acquisition

1. Veracity context-aware constraints acquisition



From <Sentence>, can we conclude that the claim is **true**?

<Answer>:

From <Sentence>, can we conclude that the claim is **false**?

<Answer>:

Constraints Knowledge

2. Veracity context-aware backgrounds acquisition

Please briefly introduce the background of drinking lemon juice. No more than 50 words. <Response>:

Please briefly introduce chemotherapy. No more than 50 words.

< Response>:

Backgrounds Knowledge

3. Veracity context-aware analogies acquisition

Given several initial fake news article and its veracity label:

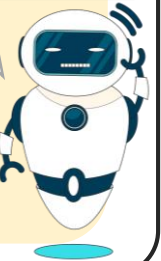
<SENT>: A recent study claims that drinking... <PRED>: Fake.

Referring <SENT>, Please provide 3~5 relevant fake or real news examples.

< Examples>:

Analogies Knowledge

Context-Aware Knowledge Querying ...



... cures cancer without chemotherapy is fake?
\n Yes
... cures cancer without chemotherapy is real?
\n No
The lemon juice, ... \n The chemotherapy ... \n
The cancer treatment is, ...

Veracity-Enriched Contexts

... drinking aloe vera juice
can eliminate diabetes, ...
Label: fake
... turmeric can completely
replace antibiotics, ...
Label : fake

Self-adaptive bootstrap prompting optimization

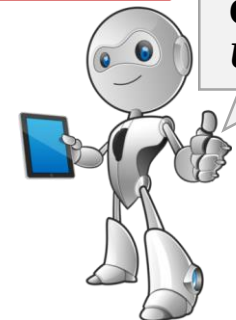


LLM

prediction

REAL
✓

FAKE
✗



Veracity-Enriched Contexts Understood !

Step 2. Veracity Context-Aware Knowledge Infusion & Fake News Detection